Option-Only Portfolio Optimization with Greek-Induced Covariance and Conditional Risk Premia

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Abstract

We formulate option-only allocation as a funded cashflow problem in which premium, short liability, margin, collateral, liquidity, and tail loss are separate quantities. A fixed-Greek expansion supplies delta, centered-gamma, and vega loadings. The allocation program uses their complete joint covariance with residual returns, structural expected returns, asymmetric costs, sample CVaR, stress limits, and assignment constraints. It converts the continuous solution to whole contracts and accepts the result only after rechecking every constraint. A legacy maximum-Sharpe specification fails because all of its VIX-enabled CPCV paths reach zero wealth in March 2020. In the 93-month development sample, the R1.1 specification with a 25 percent volatility ceiling on eight names plus VIX has a 15.8 percent annualized mean return, a 16.0 percent CAGR, a modeled-cost Sortino ratio of 2.275, and a 17.1 percent maximum drawdown. Its Sortino ratio is 2.052 in the touch-price sensitivity. Entry and round-trip quote coverage for that scenario are 94.1 and 25.5 percent. These results are retrospective. Confirmation requires 36 untouched monthly observations beginning 2026-07-31 under the registered policy, data rules, and recorded implementation hashes.

1. Introduction

An option portfolio cannot be scaled or funded like a stock portfolio. The signed premium paid today is only one cashflow. Short liability, margin, collateral, assignment, and stress loss are different quantities, and a cheap contract can carry a large normalized Greek exposure. We therefore begin with the accounting identity, derive option returns from fixed Greeks, and estimate the complete covariance of those factor exposures and their residuals. Allocation then becomes a convex, cost-aware problem with cash and hard survival limits. Whole-contract execution is a separate acceptance step.

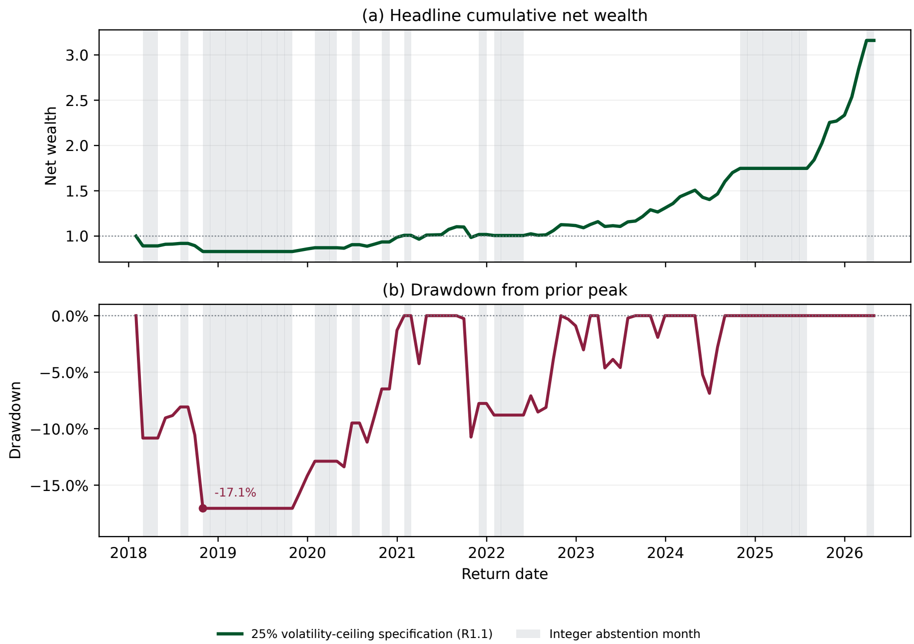

Figure 1: Retrospective development path for the eight-name plus VIX R1.1 specification. Panel (a) compounds net returns under the modeled costs. Shading marks cash selections after direct integer conversion failed a constraint. Panel (b) reports drawdown.

The empirical analysis compares three documented specifications. E1 uses a maximum-Sharpe direction and fails when all VIX-enabled CPCV wealth paths reach zero in March 2020. R1 replaces that ratio objective with net utility, includes the complete covariance, admits cash, and applies a 15 percent volatility ceiling. R1.1 keeps those rules and raises the ceiling to 25 percent.[1]

Figure 1 reports the retrospective path for the eight-name plus VIX R1.1 specification. Shading identifies months in which direct integer conversion violated a hard constraint and the rule selected cash. The implementation does not search for a substitute risky book.

Mean-variance choice supplies the starting point (Markowitz, 1952). The option return literature connects compensation to option exposure and volatility risk (Coval and Shumway, 2001; Bakshi and Kapadia, 2003; Bollerslev et al., 2009), while existing allocation work combines option returns, Greeks, and portfolio choice (Faias and Santa-Clara, 2017; Zhao and Palomar, 2018; Malamud, 2014). This paper studies funded option weights, factor-residual cross covariance, constrained net utility, and integer acceptance within one reproducible allocation rule. The failed E1 specification records the

[1]The internal codes E1, R1, and R1.1 also appear in artifact filenames and verifier keys.

consequence of evaluating a ratio without the accepted survival rule. Multiple-testing and combinatorial validation follow White (2000); Hansen (2005); Bailey et al. (2017); Bailey and López de Prado (2014).

2. Funded model and complete covariance

2.1. Funded premium weights and NAV

Let $C_{i,t} > 0$ be the mark of contract i, m_i its multiplier, $N_{i,t} \in \mathbb{Z}$ the signed number of contracts, and $W_t > 0$ portfolio NAV. The decision variable is the premium weight

$$w_{i,t} = \frac{N_{i,t} m_i C_{i,t}}{W_t}. \tag{1}$$

If A_t is the marked cash account, then

$$W_t = A_t + \sum_i N_{i,t} m_i C_{i,t}, \qquad \frac{A_t}{W_t} = 1 - \mathbf{1}^\top w_t. \tag{2}$$

For $r_{i,t+1} = (C_{i,t+1} - C_{i,t})/C_{i,t}$, a self-financing option leg contributes

$$\frac{\Delta W_{i,t+1}}{W_t} = w_{i,t} r_{i,t+1}. \tag{3}$$

The contract multiplier cancels from the return contribution because it is already in the weight.

Equation (2) is an accounting identity, not a claim that all marked cash is free collateral. A long option consumes premium. A short option produces premium but creates a marked liability whose loss can exceed the initial receipt. Broker margin is a separate reserve, and capital at risk depends on stress loss, assignment, and liquidity as well as premium. The optimizer therefore controls gross premium $\|w\|_1$, short premium $\mathbf{1}^\top w^-$, margin, collateral, and tail loss separately. Any unused admissible capital remains in cash.

2.2. Second-order option returns

Over one holding period, a second-order local expansion gives

$$\Delta C_i \simeq \Delta_i \Delta S_i + \frac{1}{2}\Gamma_i (\Delta S_i)^2 + \nu_i \Delta\sigma_i + \Theta_i \Delta t. \tag{4}$$

Writing $u_i = \Delta S_i / S_i$ and dividing by C_i produces the loadings

$$b_i^{\Delta} = \frac{\Delta_i S_i}{C_i}, \qquad b_i^{\Gamma} = \frac{\Gamma_i S_i^2}{2C_i}, \qquad b_i^{\nu} = \frac{\nu_i}{C_i}. \tag{5}$$

The factor equation centers the quadratic return so that its mean is not counted twice:

$$r_{i,t+1} = a_{i,t} + b_{i,t}^{\Delta} u_{i,t+1} + b_{i,t}^{\Gamma}\{u_{i,t+1}^2 - \mathbb{E}_t[u_{i,t+1}^2]\} + b_{i,t}^{\nu}\Delta\sigma_{i,t+1} + \varepsilon_{i,t+1}. \tag{6}$$

The intercept absorbs theta carry and the conditional mean of the quadratic term. These loadings are fixed from decision-date marks and Greeks. They are not regression slopes. The approximation omits higher-order and cross-Greek terms such as vanna and volga; their realized contribution remains in $\varepsilon_{i,t+1}$ (Black and Scholes, 1973; Merton, 1973; Shreve, 2004).

For equity options, S_i is the underlying spot. For VIX options, the price and Greek anchor is the matched VX futures contract under Black's futures-option model (Black, 1976). VIX and VVIX remain state variables, but neither replaces the traded forward in Equation (4). Expiration cashflows use the listed VRO/SOQ settlement rule.

2.3. Complete Greek and residual covariance

Stack the delta, centered-gamma, and vega innovations in f_{t+1}, and let B_t^G contain the loadings from Equation (5). In vector form,

$$r_{O,t+1} = \mu_{O,t} + B_t^G f_{t+1} + \varepsilon_{t+1}. \tag{7}$$

The fixed loadings make $\mathrm{Cov}_t(f_{t+1}, \varepsilon_{t+1})$ generally nonzero. An orthogonal-residual shortcut would therefore discard sample dependence that the Greek model did not explain.

Define the joint state and its covariance by

$$Z_{t+1} = \begin{bmatrix} f_{t+1} \\ \varepsilon_{t+1} \end{bmatrix}, \qquad S_{Z,t} = \mathrm{Var}_t(Z_{t+1}) = \begin{bmatrix} \Omega_t & \Gamma_t \\ \Gamma_t^{\top} & \Sigma_{\varepsilon,t} \end{bmatrix}. \tag{8}$$

The option covariance used by the solver is

$$\begin{aligned} \Sigma_{O,t} &= [B_t^G \; I] S_{Z,t} [B_t^G \; I]^{\top} \\ &= B_t^G \Omega_t (B_t^G)^{\top} + B_t^G \Gamma_t + \Gamma_t^{\top} (B_t^G)^{\top} + \Sigma_{\varepsilon,t}. \end{aligned} \tag{9}$$

Proposition 2.1 (Positive semidefiniteness)**.** *If $S_{Z,t} \succeq 0$, then $\Sigma_{O,t} \succeq 0$. The result does* *not require* $\Gamma_t = 0$.

Proof. For any x, set $y = [B_t^G \ I]^\top x$. Then $x^\top \Sigma_{O,t} x = y^\top S_{Z,t} y \geq 0$. □

The estimator centers the aligned factor and option observations, forms $E = R - F(B_t^G)^\top$, and stacks $Z = [F \ E]$. It standardizes each joint column, applies Ledoit-Wolf shrinkage to the standardized covariance, and rescales it (Ledoit and Wolf, 2004). With no shrinkage, Equation (9) reconstructs the aligned sample option covariance exactly, including both cross terms. The regularized estimate is symmetrized, transformed through $[B_t^G \ I]$, and given one final numerical eigenvalue floor of 10^{-10}. Appendix A.1 records the estimator algebra.

3. Estimation and optimization

3.1. Structural means and estimation error

Expected option returns use an auditable structural decomposition:

$$\widetilde{\mu}_{i,t} = \theta^*_{i,t} + b^\Delta_{i,t}\widehat{\lambda}^S_t + b^\Gamma_{i,t}\widehat{\lambda}^{var}_t + b^\nu_{i,t}\widehat{\lambda}^{vol}_t + \widehat{\lambda}^{skew}_{i,t} + \widehat{\lambda}^{RV}_{i,t}. \tag{10}$$

Theta carry, equity exposure, variance and volatility premia, skew or tail insurance, and within-underlying relative value are retained as separate ledger columns. Skew and tail terms affect the mean and named stress scenarios; they are not added as covariance loadings without an identified factor series.

The R1 and R1.1 estimator gives the historical option-return mean zero weight. It places weight 0.75 on Equation (10), then shrinks the combined estimate 75 percent toward zero:

$$\widehat{\mu}_t = (1 - 0.75)\{0 \cdot \bar{r}_t + 0.75\widetilde{\mu}_t\} = 0.1875\,\widetilde{\mu}_t. \tag{11}$$

The implementation finally clips a monthly mean only outside $[-0.35, 0.35]$. This construction avoids feeding an unregularized mean for every contract into the optimizer.

Proposition 3.1 (Breadth and mean estimation error)**.** *Suppose n contract-mean errors are sub-Gaussian with scale at most $\bar{\sigma}/\sqrt{T}$. Then*

$$\mathbb{E}\left[\max_{1\leq i\leq n} |\widehat{\mu}_i - \mu_i|\right] \leq \bar{\sigma}\sqrt{\frac{2\log(2n)}{T}}. \tag{12}$$

For every portfolio with $\|w\|_1 \leq g$, the corresponding objective perturbation is at most $g\|\widehat{\mu} - \mu\|_\infty$. The raw cross-sectional mean bound is smaller than a structural bias bound

β only when

$$n < \frac{1}{2}\exp\left(\frac{T\beta^2}{2\bar{\sigma}^2}\right). \tag{13}$$

Beyond this crossover, the stated structural bias bound is smaller than the stated raw mean bound. This comparison does not establish lower total estimation error because it does not bound the structural estimator's variance. Appendix A.2 gives the proof and the optimizer-regret bound.

3.2. The R1 feasible set

Write $w = w^+ - w^-$, where $w_i^+ = \max(w_i, 0)$ and $w_i^- = \max(-w_i, 0)$. Let $\ell_{i,t}$ be the point-in-time liquidity cap, $a_i \geq 0$ the short-margin coefficient, $\mathcal{I}_u$ the contracts on underlying u, and $\mathcal{A}_t$ the contracts that cannot be short because assignment is not admissible. The continuous feasible set includes

$$\|w\|_1 \leq 1.00, \qquad |\mathbf{1}^\top w| \leq 1.00, \tag{14}$$

$$\mathbf{1}^\top w^- \leq 0.25, \qquad |w_i| \leq \min(0.18, \ell_{i,t}), \tag{15}$$

$$\sum_{i\in\mathcal{I}_u} |w_i| \leq \begin{cases} 0.20, & u = \text{VX}, \\ 0.35, & \text{otherwise}, \end{cases} \qquad w_i \geq 0 \quad (i \in \mathcal{A}_t), \tag{16}$$

$$|(b^\beta)^\top w| \leq 3.0, \qquad |(b^{VIX\nu})^\top w| \leq 8.0, \tag{17}$$

$$R_t^{stress} w \geq -0.20\mathbf{1}, \qquad a^\top w^- \leq 0.75, \tag{18}$$

$$\mathbf{1}^\top w^+ + a^\top w^- \leq 1.00. \tag{19}$$

The liquidity cap is computed only from information available at the decision date. The VX group is labeled `VX_FRONT` in code. The beta limit is relative to SPY, and the vega limit applies to VIX exposure. The stress matrix contains the predefined scenarios. Equations (14)-(19), together with the sample CVaR constraint below, define $\mathcal{C}_t$.

Every bound is convex and $0 \in \mathcal{C}_t$. When expected edge does not pay for cost and risk, the optimizer can decline exposure. A premium receipt from a short position does not relax the margin or collateral budgets. The 15 percent annualized R1 volatility ceiling is imposed by the risk-aversion selection rule rather than inserted as another constraint in the first-stage program.

3.3. *Sample CVaR and asymmetric costs*

Let $r^{(k)}$ be training scenario k, K the number of scenarios, and

$$\kappa_t(w) = c_{+,t}^\top w^+ + c_{-,t}^\top w^-. \tag{20}$$

The vectors differ because crossing and closing a long position need not cost the same as opening and covering a short position. The scenario loss includes this predictable cost: $L_k(w) = -(r^{(k)})^\top w + \kappa_t(w)$. At $\alpha = 0.95$, the implementation uses the Rockafellar-Uryasev epigraph (Rockafellar and Uryasev, 2000)

$$\widehat{\text{CVaR}}_\alpha(w) = \min_{\eta\in\mathbb{R}} \left\{ \eta + \frac{1}{(1-\alpha)K} \sum_{k=1}^{K} [L_k(w) - \eta]_+ \right\}, \tag{21}$$
$$\widehat{\text{CVaR}}_{0.95}(w) \le 0.10.$$

The complete first-stage allocation is

$$\max_{w\in\mathcal{C}_t} \left\{ \widehat{\mu}_t^\top w - c_{+,t}^\top w^+ - c_{-,t}^\top w^- - \frac{\lambda}{2} w^\top \Sigma_{O,t} w \right\}. \tag{22}$$

This is the accepted model for both R1 and the first stage of R1.1.

Proposition 3.2 (Convexity and global optimality)**.** *For $\Sigma_{O,t} \succeq 0$ and nonnegative cost vectors, the objective in Equation* (22) *is concave and $\mathcal{C}_t$ is convex. The equivalent minimization problem is convex, so every computed optimum is global (Boyd and Vandenberghe, 2004).*

Proof. The cost κ_t and quadratic form are convex. Their negatives are concave. Absolute value, linear, quadratic-epigraph, and sample-CVaR inequalities define convex sets. □

The optimality condition is most compact in normal-cone form. If w^* solves Equation (22), then

$$0 \in -\widehat{\mu}_t + \partial\kappa_t(w^*) + \lambda\Sigma_{O,t} w^* + N_{\mathcal{C}_t}(w^*). \tag{23}$$

For one contract, the cost subgradient is $c_{+,i}$ when $w_i > 0$, $-c_{-,i}$ when $w_i < 0$, and the interval $[-c_{-,i}, c_{+,i}]$ at zero. A contract enters only when its marginal expected return clears the appropriate cost wedge after risk and active constraints are priced. The dual multipliers on CVaR, stress, liquidity, margin, collateral, and Greek limits are shadow prices for relaxing those budgets. The full multiplier form appears in Appendix A.3.

Proposition 3.3 (Existence and uniqueness)**.** *The accepted program has an optimum. If $\Sigma_{O,t} \succ 0$ and $\lambda > 0$, its portfolio weight is unique.*

Proof. The set $\mathcal{C}_t$ is nonempty because it contains zero. It is closed and bounded by the gross limit, hence compact. The objective is continuous, so it attains its maximum. Positive definiteness makes the quadratic penalty strictly convex and the maximization objective strictly concave, which gives a unique weight even if the CVaR threshold η is not unique. □

Equation (23) also characterizes cash. The all-cash solution is optimal exactly when

$$\widehat{\mu}_t \in \partial\kappa_t(0) + N_{\mathcal{C}_t}(0). \tag{24}$$

Thus a weak forecast does not require an arbitrary expected-return cutoff. It remains at zero when the asymmetric trading wedge and the normals of any binding no-short rules contain the forecast vector.

3.4. Risk aversion and the volatility ceiling

Let $\Phi(w) = \widehat{\mu}_t^\top w - \kappa_t(w)$, $q(w) = w^\top \Sigma_{O,t} w$, and let w_λ be any global solution of $\max_{w\in\mathcal{C}_t}\{\Phi(w) - \lambda q(w)/2\}$.

Proposition 3.4 (Optimized variance is nonincreasing)**.** *If $0 < \lambda_1 < \lambda_2$, then $q(w_{\lambda_2}) \leq q(w_{\lambda_1})$.*

Proof. Optimality at λ_1 and λ_2 gives

$$\begin{aligned}
\Phi(w_{\lambda_1}) - \tfrac{\lambda_1}{2} q(w_{\lambda_1}) &\geq \Phi(w_{\lambda_2}) - \tfrac{\lambda_1}{2} q(w_{\lambda_2}),\\
\Phi(w_{\lambda_2}) - \tfrac{\lambda_2}{2} q(w_{\lambda_2}) &\geq \Phi(w_{\lambda_1}) - \tfrac{\lambda_2}{2} q(w_{\lambda_1}).
\end{aligned}$$

Adding the inequalities yields $(\lambda_2 - \lambda_1)\{q(w_{\lambda_1}) - q(w_{\lambda_2})\} \geq 0$. □

Proposition 3.4 justifies a one-dimensional search. The code solves at $\lambda = 10^{-6}$. If the annualized volatility already meets the ceiling, that solution is retained. Otherwise it verifies the bracket at 10^6 and performs 18 geometric-midpoint steps, equivalent to bisection on $\log\lambda$. It keeps the smallest bracketed risk aversion whose solution is at or below 15 percent for R1 and 25 percent for R1.1.

3.5. R1.1 augmentation and whole contracts

R1.1 first solves Equation (22) with the 25 percent ceiling. Let $w^{(1)}$ be this solution and $s_i = \operatorname{sign}(w_i^{(1)})$. If $\|w^{(1)}\|_1 < 0.50$, the second stage freezes these signs. With $d_i = s_i\widehat{\mu}_i - c_i(s_i)$, it permits augmentation only on

$$\mathcal{E} = \{i : s_i \neq 0,\ d_i > 0\}, \qquad w = s \odot x, \quad x \geq 0, \quad x_i = 0\ (i \notin \mathcal{E}). \tag{25}$$

It then solves the feasibility problem

$$\begin{aligned} G^* = \max_x\ \mathbf{1}^\top x \\ \text{subject to} \quad s \odot x \in \mathcal{C}_t, \qquad x \geq 0, \qquad x_i = 0\ (i \notin \mathcal{E}), \\ (s \odot x)^\top \Sigma_{O,t}(s \odot x) \leq \left(\frac{0.25}{\sqrt{12}}\right)^2, \qquad d^\top x \geq 0, \\ \mathbf{1}^\top x \leq 0.50. \end{aligned} \tag{26}$$

Only when $G^* \geq 0.50$ does the code impose $\mathbf{1}^\top x = 0.50$ and re-solve net utility on the retained first-stage directions. Otherwise it keeps $w^{(1)}$. The target was never feasible in the historical replay: every universe and arm has a target-hit rate of 0.00, with mean gross exposure between 0.047 and 0.075.

For account NAV W_t, multiplier 100, and mark $C_{i,t}$, a continuous weight implies

$$N_{i,t} = \frac{w_{i,t}W_t}{100C_{i,t}}, \qquad \widehat{N}_{i,t} = \operatorname{sign}(N_{i,t}) \left\lfloor |N_{i,t}| \right\rfloor. \tag{27}$$

This truncation moves each leg toward cash. The execution code preserves the continuous signs, including hedge legs whose standalone edge is negative, and rechecks gross, net, short, contract, underlying, beta, vega, stress, CVaR, margin, collateral, assignment, liquidity, and volatility limits on the resulting whole-contract book. A feasible direct conversion trades. An infeasible conversion selects all cash and records the rejected book's diagnostics. The accepted policy is not a mixed-integer optimizer and does not search for a substitute integer portfolio.

3.6. Decision-time algorithm

The monthly decision follows one fixed sequence.

1. Form the eligible contract set from decision-date marks, Greeks, maturity, moneyness, volume, open interest, quote quality, assignment status, and liquidity caps. No

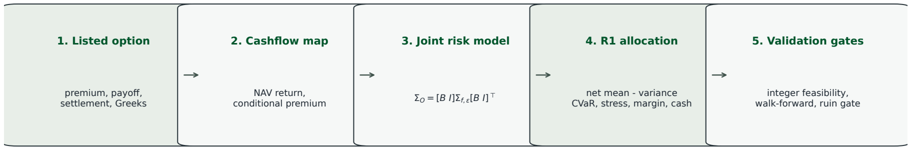

Figure 2: Theory and implementation flow. Listed contracts become NAV-normalized cashflows. The joint factor and residual covariance enters cost-aware net utility. Hard survival and execution gates apply to the whole-contract book.

later observation can alter this set.

2. On the trailing training window, compute the structural mean in Equation (11), factor panel, Greek residuals, and regularized joint covariance in Equation (9).
3. Build direction-specific costs and the training-scenario losses. Solve Equation (22) at the risk-aversion endpoints, then use the 18-step logarithmic search when the lower endpoint exceeds the policy's volatility ceiling.
4. For R1.1 only, test Equation (26). The first-stage book is retained unless the sign-constrained feasible gross reaches 0.50.
5. Convert the selected continuous weights through Equation (27). Recompute every constraint and the volatility ceiling on that exact whole-contract book. Trade it if it passes; otherwise record cash.
6. Realize the next holding-period return under the stated cost and settlement rules, write the decision and diagnostics to the ledger, and advance the training endpoint.

Estimation, optimization, and execution acceptance are separate steps. This prevents an integer repair, a later quote, or a model-price substitution from changing the portfolio that the continuous program selected.

Figure 2 summarizes the calculation. Estimation determines a continuous target; survival and execution determine whether that target is admissible. Neither a positive mean estimate nor a low estimated variance can override a zero-wealth path, a margin breach, or an infeasible integer conversion.

4. Data, cost, and validation design

The empirical panel contains listed equity options, underlying state variables, and the matched volatility instruments needed by the four specified universes. Every rebalance uses only information available by its decision date. Table 1 records the four comparison sets and their prescribed spread inputs.

Table 1: Empirical universes and baseline evidence. Prototype net is the static full-cost net Sharpe ratio used to describe the earlier development experiment.

Config	Universe	Spread input	Status	Prototype net	Naive net	Stock Markowitz
orig	8 equities	exact panel CBBO	capacity diagnostic	0.778	0.365	0.909
orig+VIX	8 equities + VIX	exact panel CBBO + inferred CBBO proxy	research candidate	1.383	-0.890	0.909
larger	56 equities	exact panel CBBO + inferred CBBO proxy	mixed	0.587	0.596	0.909
larger+VIX	56 equities + VIX	exact panel CBBO + inferred CBBO proxy	research candidate	1.628	0.293	0.909

The original universes contain eight equity names. The larger universes contain 56. The design evaluates each set with and without VIX options. Contract selection applies maturity, moneyness, mark, open-interest, volume, and quote-quality filters before optimization. The decision-date panel supplies marks and Greeks. Holding-period returns use subsequent option marks or expiry cashflows under the stated return-construction rule. VIX expiry cashflows use the listed VRO/SOQ settlement record.

The R1 replay contains 60 observed monthly returns from January 2021 through April 2026. The R1.1 replay begins in February 2018 and contains 93 observed monthly returns through April 2026. The longer R1.1 record evaluates the separately documented 25 percent ceiling; it is not an extension of the R1 headline after inspection of R1 outcomes.

The cost stack includes half-spread, fees, direction-specific slippage, short-call borrow, and funding where applicable. These transaction costs enter both selection and reported net returns. The stack is asymmetric because buying at the ask and covering a short at the ask are not mirror images of receiving the bid.

Spread evidence follows a fixed ladder. The eight equity names use exact historical panel CBBO. Rows outside that panel use an inferred CBBO proxy when no matched historical quote exists. Broad-name and VIX proxy rows are execution sensitivity, not execution truth. The licensed-quote audit later tests that distinction directly. VIX settlement requires the appropriate listed settlement record rather than a convenient close.

The baseline comparison includes ordinary underlying Markowitz and capped naive option books. The stock baseline asks whether option complexity earns its place. The capped naive book asks whether estimation adds value beyond simple option diversification under the same costs and capacity rules.

5. E1 failure and R1/R1.1 results

E1 maximized a cost-adjusted Sharpe direction after inspection of the post-2020 development window. The procedure ranks historical paths but does not make the selection prospective. Figure 3 and Table 2 retain the result as a record of the research path.

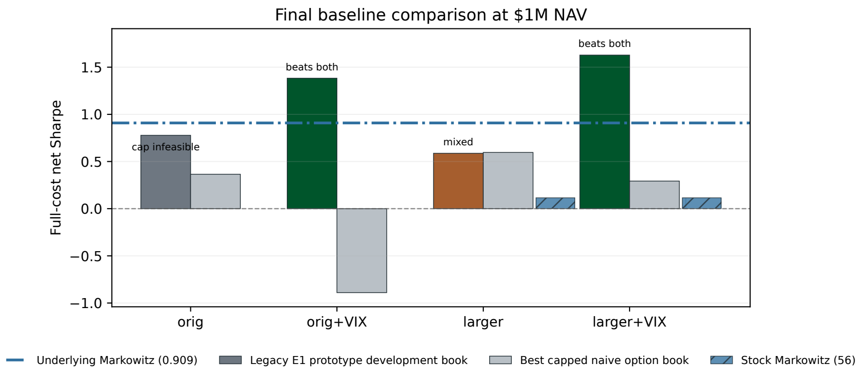

Figure 3: Static full-cost net comparison for E1, capped naive option books, and stock baselines at the stated account size. All values are retrospective development evidence.

Table 2: Structural-mean channel ablation for E1. Each row re-solves and re-scores the book after removing one channel.

Arm	orig	orig+VIX	larger	larger+VIX
Full model	0.778	1.383	0.587	1.628
No carry	0.774	0.463	0.574	0.483
No delta	0.370	1.479	0.672	1.684
No vol/VRP	0.772	1.382	0.587	1.600
No skew/tail	0.772	1.372	0.586	1.629
No relative value	0.822	1.355	0.504	1.570

The liquid-era CPCV paths reveal the survival failure. Figure 4 pivots the artifact to one column per path and compounds each path independently. Every VIX-enabled path reaches the absorption boundary in March 2020. Arithmetic return ratios after that event cannot restore wealth.

Reaching zero wealth disqualifies the prototype even though a later window compounds rapidly. Researchers selected the model after inspecting that later window, which omits the event that destroys the earlier paths.

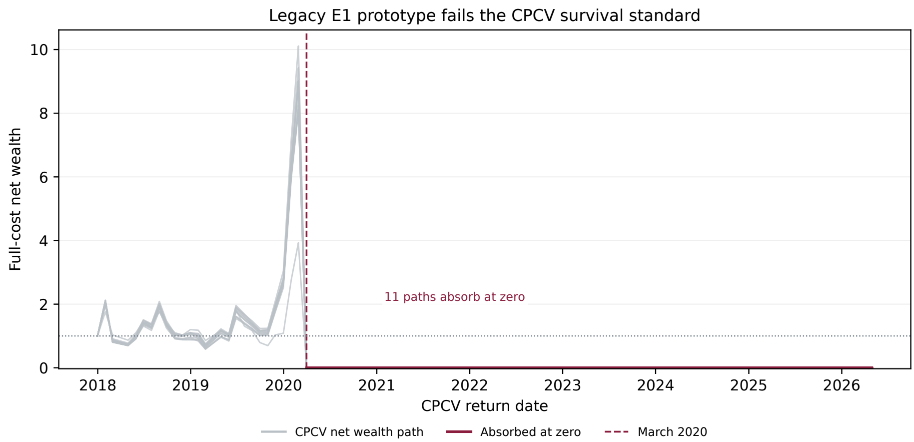

Figure 4: Full-cost CPCV wealth paths for VIX-enabled E1. All paths absorb at zero in March 2020. The maroon segment marks the absorbing state.

5.1. 15 percent volatility-ceiling specification (R1)

R1 replaces the E1 objective and acceptance rule. Its covariance retains factor-residual cross terms, and predictable costs enter utility. Cash is admissible, offsetting split-leg gross is removed, and the full survival set is checked before a whole-contract position is accepted.

Table 3: R1 monthly walk-forward development summary for 60 observed returns from January 2021 through April 2026. Terminal wealth and tail statistics describe whole-contract books under the 15 percent volatility ceiling.

Universe	Obs.	Mean gross	Terminal wealth	CAGR	Worst month	ES_{95}	Verdict
orig+VIX	60	0.049	2.814	0.230	-0.062	-0.046	survived
larger+VIX	60	0.050	2.401	0.191	-0.059	-0.048	survived
orig	60	0.032	1.220	0.041	-0.078	-0.067	survived
larger	60	0.038	1.180	0.034	-0.070	-0.061	survived

Table 3 shows that all four specified universes pass the development survival checks. The 15 percent predicted-volatility ceiling also limits return potential. That ceiling is a policy constraint. R1.1 separately evaluates a 25 percent ceiling while retaining the same zero-wealth, funding, tail-loss, and whole-contract acceptance rules.

5.2. 25 percent volatility-ceiling specification (R1.1)

R1.1 raises the predicted-volatility ceiling to 25 percent. It pursues a 50 percent gross target only along positive-edge directions that the cost-aware first stage selects. It never forces that target through an infeasible book. The continuous target converts directly to whole contracts by truncating toward cash. A feasible conversion trades. An infeasible conversion abstains.

The comparison in Figure 5 requires a qualification. The prototype's large line begins after its March 2020 zero-absorption failure, so the common window excludes the event that invalidates it. Researchers also selected the prototype after inspecting that window. The inferred CBBO proxy supplies its broad spread evidence. The R1.1 line is smaller because it follows the stated risk policy.

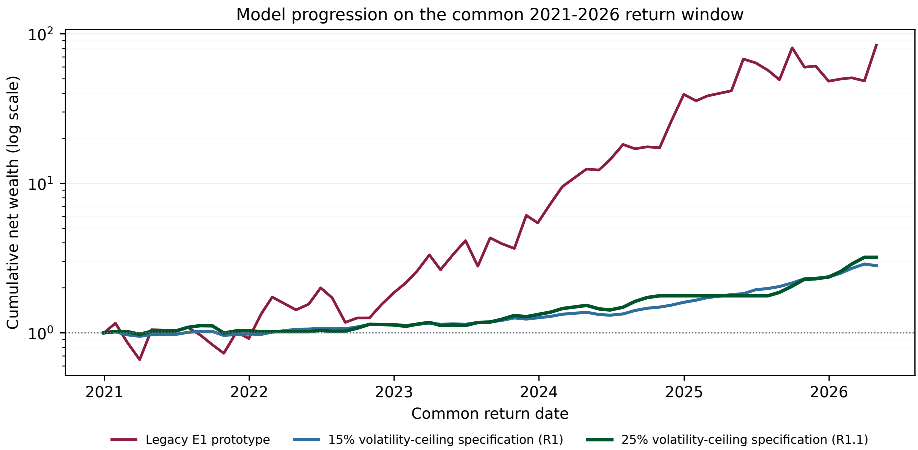

Figure 5: Common-window net wealth on a logarithmic scale. E1 reaches roughly 84 times initial wealth and R1.1 reaches about 3.2 times. The window begins after the March 2020 E1 failure, and E1 was selected after inspection of this period.

The observed gross distributions in Figure 6 sit well below the target line. The VIX-enabled books sometimes choose cash when direct whole-contract conversion is infeasible.

Table 4 keeps the base replay separate from the EGARCH diagnostic.

The VIX-40 rule reacts after an official close and uses the first executable quote after the next regular-session open. It remains unscored because the release lacks matched executable quotes for every held option; no-rule returns are never substituted.

The EGARCH overlay (Nelson, 1991) remains a diagnostic and was not selected for the candidate. It does not alter the reported base R1.1 path.

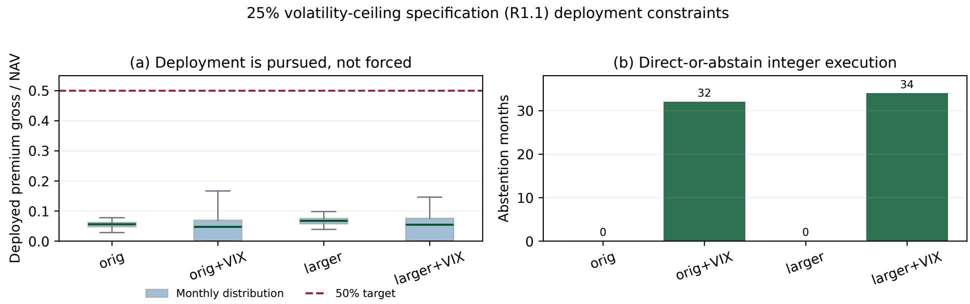

Figure 6: Deployment and integer acceptance for R1.1. Panel (a) compares realized gross premium with the target. Panel (b) counts cash abstentions under the direct conversion rule.

Table 4: Chronological R1.1 development replay for 93 observed returns from February 2018 through April 2026. Target hit is the share of whole-contract books at or above the gross target. EGARCH remains diagnostic.

Universe	Arm	Obs.	Mean gross	Target hit	Terminal	ES_{95}	Verdict
larger	EGARCH diagnostic	93	0.075	0.00	2.736	-0.113	survived
larger	R1.1 (25%)	93	0.067	0.00	2.454	-0.098	survived
larger+VIX	EGARCH diagnostic	93	0.053	0.00	2.758	-0.076	survived
larger+VIX	R1.1 (25%)	93	0.047	0.00	2.804	-0.067	survived
orig	EGARCH diagnostic	93	0.062	0.00	2.418	-0.105	survived
orig	R1.1 (25%)	93	0.055	0.00	2.368	-0.100	survived
orig+VIX	EGARCH diagnostic	93	0.058	0.00	3.018	-0.076	survived
orig+VIX	R1.1 (25%)	93	0.048	0.00	3.160	-0.077	survived

6. Observed-quote and touch-price sensitivity

The licensed-quote audit changes only spread evidence. It preserves positions, gross holding-period returns, non-spread costs, integer decisions, and survival gates. The mid scenario uses observed midpoints while retaining the stated non-spread terms. The touch scenario substitutes observed entry and exit spreads. The worst scenario uses the entry-window maximum spread and the observed exit spread.

Months without observed quotes keep the modeled cost. The audit counts these gaps rather than fabricating a quote. Reconstructing modeled cost from the recorded components leaves a maximum absolute gap below 10^{-16}. The scenario ladder therefore isolates the effect of quote-based spread evidence.

Figure 7 shows a small improvement at midpoint and lower results in the touch-price and worst-quote scenarios. The eight-name plus VIX R1.1 path does not reach zero wealth in these sensitivities. These are cost sensitivities on fixed positions, not realized execution records or backtests selected from the quote sample.

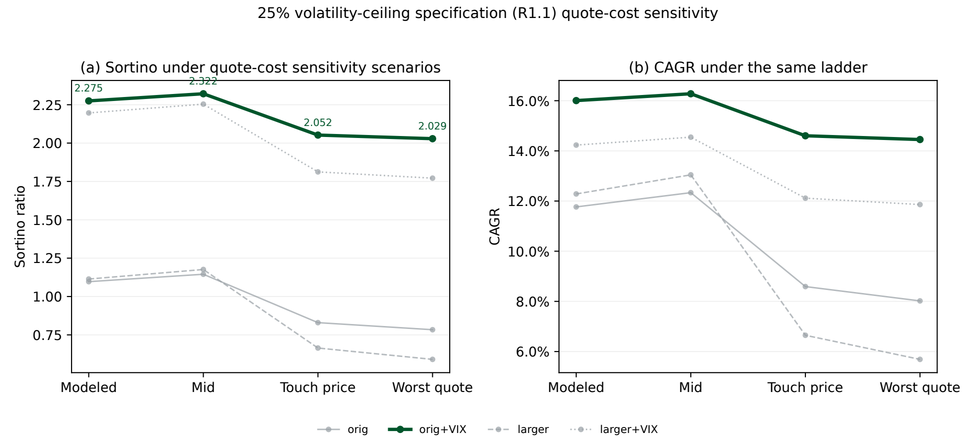

Figure 7: Quote-cost sensitivity for R1.1. The x axis moves from modeled costs through midpoint, touch-price, and worst-quote scenarios. The eight-name plus VIX configuration uses the heavier line.

6.1. Headline configuration

The spread-source distinction decides which historical result deserves emphasis. Broad-name and VIX rows lack exact panel CBBO in the historical cost stack, so those rows use the inferred proxy. The licensed audit finds wider observed broad-name spreads than the proxy assumed. Figure 8 places proxy reliance, observed-quote coverage, and spread quantiles in one view.

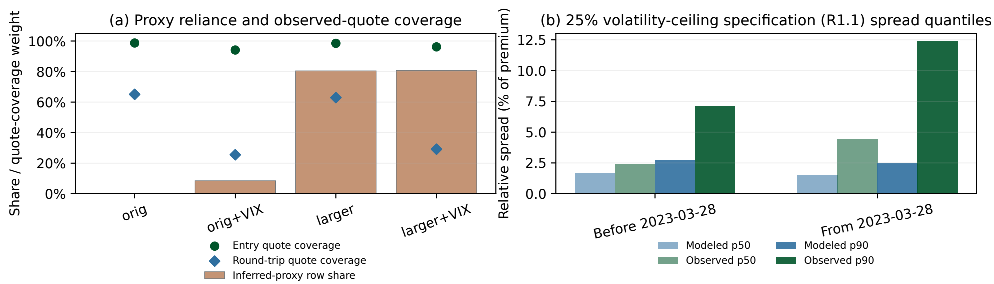

Figure 8: Proxy reliance and licensed-quote evidence. Panel (a) compares inferred-proxy row share with absolute-weighted entry and round-trip quote coverage. Panel (b) pools R1.1 positions across configurations and compares modeled and observed spread quantiles before and after the quote-schema cutover.

For the 56-name plus VIX configuration, touch-price Sortino falls from 2.197 to 1.812, a 17.5 percent decline; entry and round-trip coverage are 96.2 and 29.1 percent. For the eight-name plus VIX configuration, it falls from 2.275 to 2.052, a 9.8 percent decline;

entry and round-trip coverage are 94.1 and 25.5 percent. The eight-name plus VIX result is reported because its equity-option spread evidence is exact in the historical panel and its touch-price path does not reach zero wealth. Low round-trip coverage limits the execution interpretation. Appendix A.9 reports the full scenario, spread, and liquidity tables.

7. Robustness, capacity, and limitations

The validation suite separates path fragility from formal inference. CPCV recombines purged test blocks and preserves the zero-wealth boundary.

Monte Carlo resampling tests return ordering. Refit and reprice exercises repeat estimation and option repricing under locked cost rules. These simulations are path-risk diagnostics, not tradable histories or confirmatory observations.

Robustness status by test

Configuration	Liquid CPCV	Claim CPCV	MC resample	MC refit	Rolling OOS	PBO rank	Repriced
orig	pass	pass	pass	pass	pass	pass	mixed
orig+VIX	default	pass	pass	pass	pass	pass	mixed
larger	pass	pass	mixed	pass	pass	mixed	fail
larger+VIX	default	pass	pass	pass	pass	mixed	fail

Liquid-era CPCV default status takes precedence over Sharpe-based diagnostics.

Figure 9: Robustness status for the four E1 universes. Liquid-era CPCV marks the VIX-enabled configurations as defaulted because their paths reach zero. Claim-window and simulation results do not override that outcome.

The VIX-enabled E1 configurations have a CPCV default share of 1.000. Once wealth reaches zero, their Sharpe quantiles are not meaningful; the corrected evidence reports them as NA and assigns `fail_default`. Sharpe and Sortino ratios follow Sharpe (1966) and Sortino and Price (1994). In the liquid-era scope, full-cost net PBO is 0.1795 within the eight-name plus VIX configuration and 0.3846 when all configurations are ranked together. In the narrower post-2020 claim window, the corresponding estimates are 0.0000 and 0.4487. The scopes answer different ranking questions, and neither covers the complete research history. The incomplete R1 and R1.1 registries record lower bounds of 598 and 607 known trials.

Table 5: Formal inference for the static full-cost net prototype books over the common test window. The table reports uncertainty, probabilistic and deflated Sharpe measures, and comparisons with stock and capped naive baselines.

Config	Net Sharpe	CI lo	CI hi	PSR	DSR	dSR stock	p stock	dSR naive	p naive
orig	0.778	0.163	1.433	0.973	0.945	-0.131	0.776	0.413	0.014
orig+VIX	1.383	0.895	1.971	1.000	0.997	0.474	0.383	2.273	0.000
larger	0.587	0.029	1.173	0.925	0.745	-0.322	0.436	-0.009	0.967
larger+VIX	1.628	1.111	2.245	1.000	0.996	0.719	0.179	1.335	0.000

Rolling out-of-sample tests retrain only on earlier observations. The common ledger keeps the comparison dates fixed across universes. Table 5 shows that multiple-testing adjustment weakens the interpretation of the static prototype results but does not erase every signal. The survival failure still dominates the prototype's attractive later-window statistics.

7.1. Capacity and execution limits

Figure 10 compares deployed gross with the sum of contract liquidity caps and records the spread-source mix. The $1 million cap budget is a local diagnostic and does not justify extrapolation to $5 million NAV.

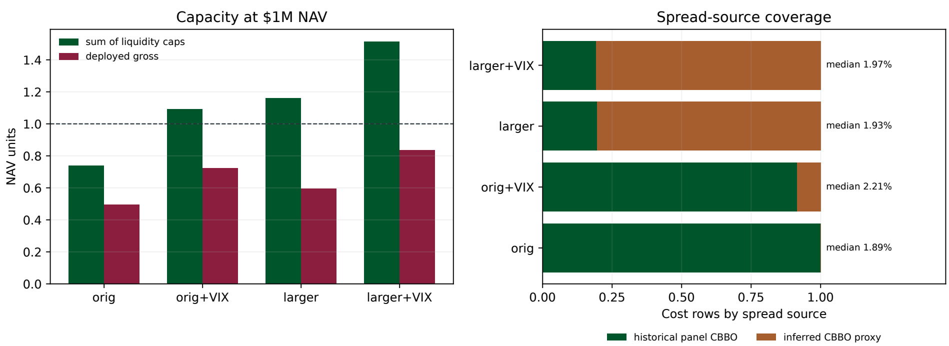

Figure 10: Capacity and spread-source evidence at the stated account size. Panel (a) compares deployed gross with the sum of liquidity caps. Panel (b) shows exact panel CBBO and inferred-proxy shares.

Multi-exchange routing could change available depth, but the paper makes no claim about routing improvement. The data contain no routed orders or realized executions. The allocator checks assignment exposure before execution. Any whole-contract breach fails closed to cash. The VIX-40 risk-off arm also remains unscored without complete event quotes. A model price cannot replace any of these missing observations after the fact.

The evidence supports one retrospective development candidate at the tested size and under the documented source hierarchy. It does not support unrestricted capacity, live execution quality, or performance outside the recorded universes.

8. Conclusion and prospective protocol

The prospective protocol freezes code, configuration, source hashes, artifact hashes, and the decision rule before the system scores a future observation. Confirmation requires 36 untouched monthly observations beginning 2026-07-31. A missing quote, failed integer conversion, or survival breach keeps its original status in the ledger. No later return can change that status.

E1 compounds rapidly in the later window but fails when its March 2020 CPCV paths reach zero wealth. R1 and R1.1 admit cash, retain the complete covariance, and enforce the stated survival limits. The eight-name plus VIX R1.1 path remains solvent in the touch-price cost sensitivity. It is a retrospective candidate, not a confirmatory result.

A. Technical appendix

This appendix gives the estimator algebra, proofs, legacy optimization geometry, and the equation-to-code map. The empirical tables and two detailed validation figures follow the mathematical material.

A.1. Joint covariance estimator

Let $F \in \mathbb{R}^{T\times k}$ and $R \in \mathbb{R}^{T\times n}$ be centered, date-aligned factor and option-return matrices. The decision-date Greek matrix is $B^G \in \mathbb{R}^{n\times k}$. The residual matrix is not an OLS residual matrix:

$$E = R - F(B^G)^\top. \tag{28}$$

Because B^G comes from marks and Greeks, no normal equation sets $F^\top E$ to zero. The cross covariance $\widehat{\Gamma} = F^\top E/(T-1)$ is therefore estimated and retained.

Set $Z = [F\ E]$ and let $D = \operatorname{diag}(s_1, \dots, s_{k+n})$ contain its sample standard deviations. A zero-scale column uses a safe unit divisor during standardization and returns to zero when rescaled. With $\widetilde{Z} = ZD^{-1}$, the standardized Ledoit-Wolf estimate has the form

$$\widehat{S}_{LW} = (1-\widehat{\rho})\frac{\widetilde{Z}^\top \widetilde{Z}}{T} + \widehat{\rho}\widehat{\tau}I. \tag{29}$$

The implementation uses the finite-sample shrinkage coefficient returned by the standard Ledoit-Wolf estimator, then rescales:

$$\widehat{S}_Z = D\widehat{S}_{LW}D = \begin{bmatrix} \widehat{\Omega} & \widehat{\Gamma} \\ \widehat{\Gamma}^\top & \widehat{\Sigma}_\varepsilon \end{bmatrix}. \tag{30}$$

Congruence by D preserves positive semidefiniteness. A second congruence by $H = [B^G\ I]$ gives $\widetilde{\Sigma}_O = H\widehat{S}_Z H^\top \succeq 0$. Floating-point asymmetry is removed before the eigendecomposition

$$\widehat{\Sigma}_O = Q\,\mathrm{diag}\{\max(\xi_j, 10^{-10})\}Q^\top, \qquad \frac{\widetilde{\Sigma}_O + \widetilde{\Sigma}_O^\top}{2} = Q\,\mathrm{diag}(\xi_j)Q^\top. \tag{31}$$

This final operation is a numerical floor, not another statistical shrinkage model.

The unregularized identity is useful for testing. If $\widehat{S}_Z = Z^\top Z/(T-1)$, then

$$\begin{aligned} H\widehat{S}_Z H^\top &= \frac{1}{T-1}(ZH^\top)^\top(ZH^\top) \\ &= \frac{1}{T-1}\{F(B^G)^\top + E\}^\top\{F(B^G)^\top + E\} = \frac{R^\top R}{T-1}. \end{aligned} \tag{32}$$

Thus both cross terms are necessary for exact reconstruction. Setting $\widehat{\Gamma} = 0$ would describe a different estimator.

A.2. Breadth and mean error

Let $e_i = \widehat{\mu}_i - \mu_i$, and suppose $\mathbb{E}[\exp(se_i)] \le \exp(s^2\bar{\sigma}^2/(2T))$ for every i. Applying the same bound to $-e_i$ gives, for $s > 0$,

$$\begin{aligned} \mathbb{E}\left[\max_i |e_i|\right] &\le \frac{1}{s}\log \mathbb{E}\left[\sum_{i=1}^n \{e^{se_i} + e^{-se_i}\}\right] \\ &\le \frac{\log(2n)}{s} + \frac{s\bar{\sigma}^2}{2T}. \end{aligned} \tag{33}$$

Minimizing the right side at $s = \sqrt{2T\log(2n)}/\bar{\sigma}$ proves Equation (12).

For any $\|w\|_1 \le g$, Holder's inequality gives

$$|e^\top w| \le \|e\|_\infty \|w\|_1 \le g\|e\|_\infty. \tag{34}$$

Let w^* maximize the true cost-aware utility and $\widehat{w}$ maximize the same utility with $\widehat{\mu}$. Adding and subtracting estimated utility, then using the two optimality inequalities, yields

$$J_\mu(w^*) - J_\mu(\widehat{w}) \le 2g\|\widehat{\mu} - \mu\|_\infty. \tag{35}$$

If a structural estimator has uniform bias at most β, the raw cross-sectional mean bound is smaller only while

$$n < \frac{1}{2}\exp\left(\frac{T\beta^2}{2\bar{\sigma}^2}\right). \tag{36}$$

Beyond this crossover, the stated structural bias bound is smaller than the stated raw mean bound. The comparison does not rank the estimators by total error because it does not bound the structural estimator's variance. It only shows how the raw mean bound changes when contract breadth rises without a longer training sample.

A.3. Optimality conditions and shadow prices

Write every feasible-set inequality as $g_j(w, \eta) \le 0$, including the sample-CVaR epigraph, and retain linear equalities as $h_l(w) = 0$. The convex minimization form of Equation (22) has Lagrangian

$$\mathcal{L}(w, \eta, \rho, \zeta) = -\widehat{\mu}^\top w + \kappa(w) + \frac{\lambda}{2} w^\top \Sigma_O w + \sum_j \rho_j g_j(w, \eta) + \sum_l \zeta_l h_l(w), \quad (37)$$

where $\rho_j \ge 0$. Under the usual relative-interior qualification, the KKT system is

$$0 \in -\widehat{\mu} + \partial\kappa(w^*) + \lambda\Sigma_O w^* + \sum_j \rho_j^* \partial_w g_j(w^*, \eta^*) + \sum_l \zeta_l^* \nabla h_l(w^*), \quad (38)$$

$$0 \in \sum_j \rho_j^* \partial_\eta g_j(w^*, \eta^*), \quad (39)$$

$$g_j(w^*, \eta^*) \le 0, \qquad h_l(w^*) = 0, \qquad \rho_j^* \ge 0,$$
$$\rho_j^* g_j(w^*, \eta^*) = 0. \quad (40)$$

These conditions are sufficient because the problem is convex. They reduce to Equation (23) when the individual constraint normals are collected in $N_{\mathcal{C}}(w^*)$.

The multiplier on a binding right-hand-side budget measures the local gain in optimized utility from relaxing that budget, subject to the regularity conditions for sensitivity analysis. A high CVaR multiplier identifies scarce tail capacity. The same reading applies to stress, liquidity, margin, collateral, beta, and vega multipliers. Zero multipliers on slack limits prevent a post hoc story in which every listed constraint is said to drive the portfolio.

For completeness, the sample-CVaR term can be written without positive-part notation. Introduce $u_k \ge 0$ and impose

$$u_k \ge L_k(w) - \eta \quad (k = 1, \ldots, K), \qquad \eta + \frac{1}{(1-\alpha)K} \sum_{k=1}^{K} u_k \le L^{CVaR}. \quad (41)$$

For fixed (w, η), minimizing the left side sets $u_k = [L_k(w) - \eta]_+$. The auxiliary formulation is therefore equivalent to Equation (21). Since $L_k(w)$ is convex after asymmetric costs, these epigraph inequalities remain convex. The solver handles them together with the quadratic risk term through its conic interface.

A.4. Finite scale and conservative changes

Lemma A.1 (No free option scale)**.** *For any nonzero premium-weight direction d, the feasible ray $\{ad : a \geq 0\} \cap \mathcal{C}$ is bounded.*

Proof. The gross-premium constraint requires $a\|d\|_1 \leq 1$. Since $d \neq 0$, $a \leq 1/\|d\|_1 < \infty$. A direction with $d = 0$ has no option premium, payoff, or Greek exposure and is economically empty. The additional stress, margin, assignment, and liquidity limits can only shorten the feasible ray. □

Define $V(\mathcal{C}, c_+, c_-)$ as the optimized value of Equation (22). If $\mathcal{C}_1 \subseteq \mathcal{C}_0$, $c_{+,1} \geq c_{+,0}$, and $c_{-,1} \geq c_{-,0}$ componentwise, then

$$V(\mathcal{C}_1, c_{+,1}, c_{-,1}) \leq V(\mathcal{C}_0, c_{+,0}, c_{-,0}). \tag{42}$$

Every point feasible under the tighter rules was feasible before, and its new cost is no smaller. This ordering concerns optimized utility, not realized return. It is the correct formal statement behind the spread and capacity sensitivity checks.

A.5. Legacy maximum-Sharpe reduction

The legacy E1 specification used a ratio objective that is not the accepted R1 or R1.1 program. For $\Sigma \succ 0$, excess-mean vector a, and convex cone K, consider

$$\max_{w \in K \setminus \{0\}} \frac{a^\top w}{\sqrt{w^\top \Sigma w}}. \tag{43}$$

The ratio is homogeneous of degree zero. Any direction with positive numerator can be normalized to variance one, which gives the conic problem

$$\max_{y \in K} \; a^\top y \quad \text{subject to} \quad \|\Sigma^{1/2} y\|_2 \leq 1. \tag{44}$$

Conversely, any nonzero optimizer of Equation (44) supplies an optimal ratio direction. Without constraints, that direction is proportional to $\Sigma^{-1} a$.

This reduction does not choose an economically valid option scale. It also does not represent the asymmetric piecewise-linear costs, admissible cash, sample CVaR, or whole-contract acceptance rule in Equation (22). E1 used it for selection; R1 and R1.1 do not.

A.6. R1.1 geometry and integer acceptance

Proposition A.2 (Convex sign-restricted augmentation)**.** *After the first-stage signs s are fixed, the feasible set in Equation* (26) *is convex, and gross premium is $\mathbf{1}^\top x$. Therefore the maximum-feasible-gross problem and the conditional 50 percent utility problem have global solutions whenever they are feasible.*

Proof. The map $x \mapsto s \odot x$ is linear. The inverse image of $\mathcal{C}_t$ under this map is convex. Nonnegativity, eligibility, total edge, and gross limits are linear. The variance cap is a convex quadratic inequality because $\Sigma_O \succeq 0$. Finally, $|s_i x_i| = x_i$ on every retained nonzero sign, so $\|s \odot x\|_1 = \mathbf{1}^\top x$. Both objectives are concave or linear on this set. $\square$

The feasibility stage matters because an equality such as $\mathbf{1}^\top x = 0.50$ can make an otherwise valid convex program infeasible. Solving for G^* first distinguishes an economic target from an available risk and liquidity budget. In the historical replay, $G^* < 0.50$ at every decision, so the equality branch was never entered.

Truncating each contract count toward zero reduces the premium magnitude of every leg, but it does not prove portfolio feasibility. A smaller hedge can increase net beta, vega, or scenario loss because those constraints depend on signed sums. The direct book must therefore pass the same diagnostic function as the continuous target. This is also why the policy does not drop a negative standalone-edge hedge during integer conversion. The accepted action is the exact truncation when feasible and cash otherwise.

The size of the discretization error is explicit. Let $u_i = 100C_i/W$ be one contract expressed as a NAV weight and let $\delta = \widehat{w} - w$. Truncation on a fixed sign gives

$$|\delta_i| < u_i \quad \text{for every active leg,} \qquad \|\delta\|_1 < \sum_{i \in \mathcal{I}(w)} u_i. \tag{45}$$

Consequently,

$$|\widehat{\mu}^\top \delta| \le \|\widehat{\mu}\|_\infty \|\delta\|_1, \tag{46}$$

$$|\kappa(\widehat{w}) - \kappa(w)| \le \max_i (c_{+,i}, c_{-,i}) \|\delta\|_1, \tag{47}$$

$$|\widehat{w}^\top \Sigma_O \widehat{w} - w^\top \Sigma_O w| \le \|\Sigma_O\|_2 \|\delta\|_2 (\|\widehat{w}\|_2 + \|w\|_2). \tag{48}$$

These bounds shrink with account NAV and grow with option price. They measure discretization error, but signed exposure cancellations can still change after truncation. Exact constraint rechecking remains necessary.

A.7. Purging and path construction

Lemma A.3 (Holding-window purge)**.** *Suppose label j uses information from a holding interval $[t_j, t_j+h_j]$. Removing every training label whose interval intersects a test interval prevents a realized return used in a test label from also entering a training label.*

Proof. If one realized return entered both labels, its date would belong to both holding intervals, so the intervals would intersect. The purge rule would have removed that training label, a contradiction. □

The implementation also embargoes observations adjacent to each test block and checks the realized calendar gap on both sides. Purging prevents label-window overlap; it does not assert statistical independence of nonoverlapping market returns. Recombined CPCV paths retain time order within each test block. Wealth stays at zero after any return of -100 percent or less, so later arithmetic ratios cannot revive an absorbed path.

A.8. Equation-to-code map

The public package implements the pricing, covariance, allocation, policy, and inference objects used in the paper. Historical factor construction and conditional-mean estimation depended on licensed feature stores. Those stages are represented by source hashes and derived evidence, not by a public regeneration pipeline.

Object	Public implementation or boundary	Role
Pricing and Greek loadings	`bs_price`, `bs_greeks`, `greek_exposure_frame`, `risk_exposure_frame`	Prices the option and constructs the factor and policy-risk loadings.
Joint covariance	`estimate_greek_joint_moments`	Estimates Equations (28)-(30); the model constructor applies Equation (9) and the eigenvalue floor.
Historical factors and means	Recorded source hashes and derived evidence	The public model accepts prepared expected returns and joint moments; it does not reconstruct the licensed feature stores behind Equations (10)-(11).
Costs and assignment inputs	`OptimizationCostSpec`	Validates prepared c_+, c_-, short-margin coefficients, and admissible-short flags.
R1 constraints	`r1_constraints`, `_net_utility_constraints`	Constructs and applies Equations (14)-(21).
R1 objective and risk search	`solve_net_utility`	Solves Equation (22) and performs the 18-step log bisection.
Independent diagnostics	`validate_portfolio`	Recomputes tail, stress, funding, assignment, and liquidity breaches.
R1.1 augmentation	`solve_r11_net_utility`	Implements Equations (25)-(26).
Whole-contract acceptance	`integerize_or_cash`	Applies Equation (27), rechecks the book, and selects direct execution or cash.
Inference	`probabilistic_sharpe_ratio`, `deflated_sharpe_ratio`	Computes the finite-sample Sharpe statistics reported with the retained evidence.

A.9. Licensed-quote audit tables

Table 7 contains the complete scenario ladder. The audit aggregates coverage by absolute portfolio weight. Missing quote legs retain modeled spread cost, and the coverage denominator still counts them. Table 6 records the source mix, while Tables 8 and 9 report the spread and entry-date liquidity checks.

Table 6: Historical spread-source ladder by universe. The counts describe cost-evidence rows, not orders or realized executions.

Configuration	Exact CBBO rows	Proxy rows	Proxy share	Median spread
orig	5777	0	0.000	0.019
orig+VIX	5777	536	0.085	0.022
larger	5777	23740	0.804	0.019
larger+VIX	5777	24276	0.808	0.020

Table 7: R1.1 CAGR, Sortino ratio, drawdown, and quote coverage under the modeled, midpoint, touch-price, and worst-quote scenarios.

Book	Cost scenario	CAGR	Sortino	MaxDD	Entry %	Round trip %
larger	modeled	0.123	1.114	-0.394	98.472	62.902
larger	midpoint	0.130	1.176	-0.391	98.472	62.902
larger	touch price	0.066	0.664	-0.413	98.472	62.902
larger	worst quote	0.057	0.590	-0.418	98.472	62.902
larger+VIX	modeled	0.142	2.197	-0.121	96.157	29.134
larger+VIX	midpoint	0.145	2.254	-0.121	96.157	29.134
larger+VIX	touch price	0.121	1.812	-0.128	96.157	29.134
larger+VIX	worst quote	0.119	1.771	-0.129	96.157	29.134
orig	modeled	0.118	1.097	-0.381	98.696	65.061
orig	midpoint	0.123	1.145	-0.378	98.696	65.061
orig	touch price	0.086	0.830	-0.389	98.696	65.061
orig	worst quote	0.080	0.784	-0.390	98.696	65.061
orig+VIX	modeled	0.160	2.275	-0.171	94.096	25.504
orig+VIX	midpoint	0.163	2.322	-0.167	94.096	25.504
orig+VIX	touch price	0.146	2.052	-0.185	94.096	25.504
orig+VIX	worst quote	0.145	2.029	-0.186	94.096	25.504

Table 8: Modeled and observed relative-spread quantiles by policy and quote regime.

Arm	Regime	Source	p25	p50	p75	p90
R1	post_2023_03_28	modeled	0.012	0.015	0.017	0.030
R1	post_2023_03_28	observed	0.028	0.046	0.078	0.133
R1	pre_2023_03_28	modeled	0.011	0.016	0.017	0.034
R1	pre_2023_03_28	observed	0.013	0.026	0.049	0.081
R1.1	post_2023_03_28	modeled	0.011	0.015	0.016	0.024
R1.1	post_2023_03_28	observed	0.026	0.044	0.073	0.124
R1.1	pre_2023_03_28	modeled	0.013	0.017	0.018	0.027
R1.1	pre_2023_03_28	observed	0.014	0.024	0.042	0.071

Table 9: Median entry-date participation and threshold breaches in the licensed-quote audit.

Policy	Vol. sum p50	Vol. max p50	OI p50	Opt. fail	Vol. fail	OI fail
R1	0.002	0.006	0.000	83	27	54
R1.1	0.004	0.011	0.001	186	57	140

A.10. Integer, concentration, and validation tables

The integer ledger in Table 10 reports accepted books and cash outcomes separately. Tables 11 and 12 retain E1 concentration and validation evidence.

Table 10: Whole-contract direct conversion and cash outcomes for R1.1 base and diagnostic books.

Method	Candidates	Feasible	Selected
Cash abstention	744	744	123
Direct truncation	744	621	621

Table 11: E1 contract breadth, concentration, deployed gross, and capacity budget at the stated account size.

Config	Candidates	Active	Top 5 share	Deployed gross	At cap share	Cap budget
orig	49	47	0.421	0.497	0.064	0.739
orig+VIX	54	54	0.324	0.723	0.093	1.093
larger	262	243	0.233	0.594	0.078	1.160
larger+VIX	267	253	0.296	0.836	0.083	1.515

Table 12: E1 CPCV Sharpe quantiles, default shares, and relative-path evidence for the liquid and claim windows. Liquid-window Sharpe is NA for configurations whose paths reach zero wealth.

Config	Liquid SR p05	Liquid SR p50	Default share	Claim SR p05	Claim SR p50	Rel. liquid p05	Rel. claim p05
orig	0.860	0.908	0.000	0.437	0.490	-0.561	-0.125
orig+VIX	NA	NA	1.000	1.003	1.062	0.214	2.257
larger	0.766	0.802	0.000	0.660	0.706	-0.265	-0.152
larger+VIX	NA	NA	1.000	1.445	1.474	0.103	1.501

A.11. Relocated path figures

Figures 11 and 12 retain the detailed path and distribution views used by the verification suite.

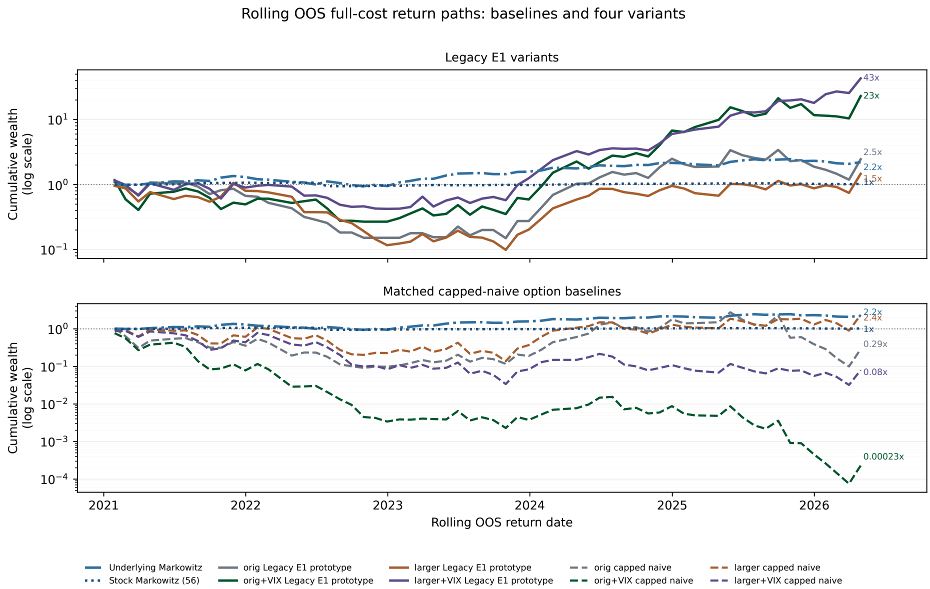

Figure 11: Rolling out-of-sample full-cost return paths for stock baselines, E1 universes, and capped naive option books.

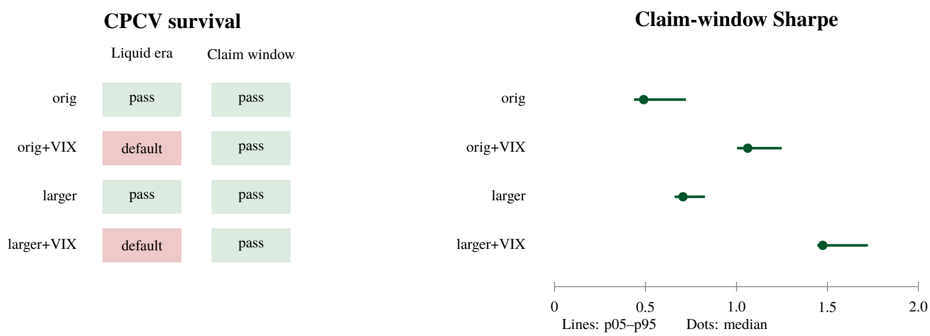

Figure 12: CPCV validation for the four E1 universes. The left panel reports survival in the liquid and claim windows. The right panel shows claim-window Sharpe quantiles; no Sharpe result is reported for a liquid-window path that defaults.

Data availability statement

The public repository contains the mathematical allocation code, the paper, and compact derived evidence. It does not redistribute raw quote, trade, contract, or licensed market data, and it does not promise controlled reviewer access. Aggregate portfolio-month results and recorded source hashes allow the reported summaries to be checked, but they cannot reconstruct unavailable feature stores or historical quotes.

The quote audit checks entry-window costs and decision marks for held positions against licensed consolidated quotes. It contains neither routed orders nor realized executions. The baseline equity-option spread rows use historical panel CBBO. Broad-name and VIX rows without matched CBBO use the documented point-in-time proxy and remain sensitivity evidence. VIX option returns follow the listed VRO/SOQ settlement convention.

References

D. H. Bailey and M. López de Prado. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management*, 40(5):94-107, 2014.

D. H. Bailey, J. M. Borwein, M. López de Prado, and Q. J. Zhu. The probability of backtest overfitting. *Journal of Computational Finance*, 20(4):39-69, 2017.

G. Bakshi and N. Kapadia. Delta-hedged gains and the negative market volatility risk premium. *The Review of Financial Studies*, 16(2):527-566, 2003.

F. Black. The pricing of commodity contracts. *Journal of Financial Economics*, 3(1-2): 167-179, 1976.

F. Black and M. Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.

T. Bollerslev, G. Tauchen, and H. Zhou. Expected stock returns and variance risk premia. *The Review of Financial Studies*, 22(11):4463-4492, 2009.

S. Boyd and L. Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004. URL https://web.stanford.edu/~boyd/cvxbook/.

J. D. Coval and T. Shumway. Expected option returns. *The Journal of Finance*, 56(3): 983-1009, 2001.

J. A. Faias and P. Santa-Clara. Optimal option portfolios. *Journal of Financial and Quantitative Analysis*, 52(1):277-303, 2017.

P. R. Hansen. A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365-380, 2005.

O. Ledoit and M. Wolf. Honey, I shrunk the sample covariance matrix. *The Journal of Portfolio Management*, 30(4):110-119, 2004.

S. Malamud. Portfolio selection with options and transaction costs. *Swiss Finance Institute Research Paper Series*, 14(08), 2014. doi: 10.2139/ssrn.2397760.

H. Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77-91, 1952.

R. C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141–183, 1973.

D. B. Nelson. Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2):347–370, 1991.

R. T. Rockafellar and S. Uryasev. Optimization of conditional value-at-risk. *The Journal of Risk*, 2(3):21-41, 2000. URL `https://sites.math.washington.edu/~rtr/papers/rtr179-CVaR1.pdf`.

W. F. Sharpe. Mutual fund performance. *The Journal of Business*, 39(1):119–138, 1966.

S. E. Shreve. *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer, 2004.

F. A. Sortino and L. N. Price. Performance measurement in a downside risk framework. *The Journal of Investing*, 3(3):59–64, 1994.

H. White. A reality check for data snooping. *Econometrica*, 68(5):1097-1126, 2000.

L. Zhao and D. P. Palomar. A Markowitz portfolio approach to options trading. *IEEE Transactions on Signal Processing*, 66(16):4223-4238, 2018. doi: 10.1109/TSP.2018.2849733.